Project Design Phase

Proposed Solution

Project Title	India's Agricultural Crop Production Analysis
Domain	Agriculture / Data Analytics & Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Historical Indian Crop Production Data, 1997–2021
Prepared By	Divya Kauluri (Team Project — completed individually; the team lead and other assigned members did not contribute to the work)
Mentor	N/A — Individual Submission
GitHub Repository	https://github.com/divyakauluri-spec/India-s-agricultural-crop-production-analysis
Tableau Public Link	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject
Date	02/07/2026

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Parameter	Description
Problem Statement	India's agricultural stakeholders — farmers, policy analysts, and investors — lack a unified, interactive visual analytics tool to explore 20+ years of crop production data by state, season, crop type, and year. This leads to suboptimal farming decisions, ineffective policy targeting, and higher investment risk.
Idea / Solution Description	An interactive Tableau-based BI dashboard solution that transforms raw government crop production data (1997–2021) into nine visualizations combined into four role-specific dashboards and a Tableau Story, published on Tableau Public for free browser access.
Novelty / Uniqueness	Role-specific dashboards tailored to three distinct user personas rather than a single generic view. The Tableau Story adds a narrative dimension that guides non-technical users through complex data. The Yield = Production/Area calculated field adds an analytical layer not present in the raw dataset.
Social Impact / Customer Satisfaction	Farmers gain data-driven crop selection guidance, reducing income risk. Policy makers gain visibility into low-yield regions, enabling better resource allocation. Investors gain a clear view of high-growth agricultural sectors, reducing investment risk.
Business Model	Individual capstone academic project delivered as a public Tableau dashboard. Future commercial potential: a paid subscription analytics platform for agribusiness firms, or a government-contracted BI solution for state agricultural departments.
Scalability	New years of data can be appended to the CSV source and the workbook refreshed. Additional crop types, states, or KPIs (e.g., rainfall, MSP prices) can be added as new calculated fields without redesigning the dashboard architecture.